LIAQUAT COLLEGE OF NURSING FEMALE, JAMSHORO

INTIMATE HYGIENE PRACTICES AMONG WOMEN IN JAMSHORO: A COMPARATIVE STUDY BASED ON EDUCATION LEVEL

A Descriptive Cross-Sectional Study

SUBMITTED BY

Atiya Hassan d/o Abul Hassan

BS Nursing (Batch 2022–2025)

SUPERVISOR

Mr. Muhammad Zakarya

BSN, MSN(In Progress)

LIAQUAT COLLEGE OF NURSING FEMALE, JAMSHORO

DECLARATION

I, **Atiya Hassan**, hereby declare that the dissertation titled *"Intimate Hygiene Practices Among Women in Jamshoro: A Comparative Study Based on Education Level"*, which I have submitted, is entirely the result of my own work. Every effort has been made to indicate this clearly, with due reference to the literature. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any other degree or diploma.

This work was carried out under the supervision and guidance of **Mr. Muhammad Zakarya**, Nursing Instructor, Liaquat College of Nursing Female, LUMHS, Jamshoro.

Place: Liaquat College Of Nursing Female, Jamshoro

Date: _____

Name: Atiya Hassan

Signature: _____

CERTIFICATE

This is to certify that **Atiya Hassan** has completed this research work titled *"Intimate Hygiene Practices Among Women in Jamshoro: A Comparative Study Based on Education Level"* as a research scholar under my supervision and guidance, for the partial fulfillment of the requirements for the degree of Bachelor of Science in Nursing at Liaquat College of Nursing Female, Jamshoro.

The research work is original, and the student has demonstrated satisfactory scholarly aptitude in conducting and presenting this study.

Supervisor's Name:

Muhammad Zakarya

BSN, MSN(In Progress), Nursing Instructor, Liaquat College of Nursing Female, LUMHS, Jamshoro

Signature: _____

Date: _____

DEDICATION

*In the name of **ALMIGHTY ALLAH**, the Most Gracious, the Most Merciful, whose infinite blessings and divine guidance illuminated every step of this journey.*

*To my beloved **parents**, whose endless sacrifices, heartfelt prayers, and unconditional love have been the foundation of everything I have achieved. Whatever I am today is because of you.*

*And to every **woman** who deserves access to proper health information and the dignity of good hygiene.*

ACKNOWLEDGEMENT

All praises and thanks are due to **Almighty Allah**, the Most Beneficent and the Most Merciful, who bestowed upon me the wisdom, strength, and determination to complete this research work successfully.

I would like to express my deepest and most heartfelt gratitude to my esteemed supervisor, **Mr. Muhammad Zakarya**, whose exceptional mentorship, profound knowledge, and tireless dedication have been the guiding light throughout this entire research journey. His scholarly expertise, insightful feedback, and remarkable patience at every stage of this thesis from the initial concept to the final manuscript were truly invaluable. It was his constant encouragement, constructive criticism, and unwavering belief in my abilities that made the successful completion of this work possible. I consider myself truly fortunate to have worked under the supervision of such a distinguished and inspiring mentor.

I owe a deep debt of gratitude to my loving **parents**, whose prayers, sacrifices, and unwavering moral support have been my greatest strength. I am equally grateful to all the **participants** who generously gave their time and candidly shared their experiences during data collection, without whom this study would not have been possible.

Special thanks to **Mr. Afzal Khan**, BS Computer Science (University of Sindh, Jamshoro), for his valuable technical assistance with statistical analysis and manuscript preparation.

Lastly, I am sincerely thankful to all my dear **friends** and **well-wishers** who stood by me, encouraged me, and believed in me at every step. May Allah bless you all.

ABSTRACT

Background: Women's intimate hygiene especially during menstruation is shaped by how much education they have, what their mothers told them, and what the local culture allows them to say out loud. In Sindh, Pakistan, very little local data exists on this topic, which means healthcare planners are working with guesswork rather than facts.

Objective: This study set out to describe what women in Jamshoro actually know, believe, and do regarding intimate hygiene and to see whether education level makes a measurable difference in their practices.

Methods: We used a descriptive cross-sectional design. One hundred women filled out a structured online questionnaire covering four areas: demographics, knowledge, attitudes, and practices. One hundred responses were valid and included in the analysis (one excluded due to a data recording error).

Results: Most participants were between 20 and 25 years old (61.6%), and the majority were students (63.0%). Knowledge of menstruation as a normal process was high 96.0% got this right. But practice was another story. Four in ten women (41.0%) believed they should not bathe during their period. Around half (50.0%) reported using vaginal douching at least sometimes. Mothers were the main source of hygiene knowledge for 51.0% of participants. And 74.0% agreed that religious activities should be avoided during menstruation.

Conclusion: Basic awareness about menstruation exists, but actual hygiene behavior has serious gaps particularly avoidance of bathing, douching, and infrequent pad changes. These habits carry real infection risks. The findings point clearly toward the need for school-level and community-based health education, especially targeting the households where mothers pass on outdated practices.

Keywords: Intimate hygiene, menstrual hygiene management, women's health, education, knowledge attitudes and practices, Jamshoro, Sindh, Pakistan

TABLE OF CONTENTS

Abstract

1. Introduction

    1.1 Gaps in Current Research

    1.2 Study Objectives

2. Review of Related Literature

    2.1 Synthesis of Evidence

3. Materials and Methods

    3.1 Study Design

    3.2 Study Setting and Duration

    3.3 Sample Size

    3.4 Data Collection Instrument

    3.5 Ethical Considerations

    3.6 Data Analysis

4. Results

    4.1 Demographic Characteristics

    4.2 Knowledge Assessment

    4.3 Attitudes

    4.4 Practices

5. Discussion

6. Conclusion and Recommendations

References

1. INTRODUCTION

Women's intimate hygiene which covers how they manage menstruation, clean themselves, and seek care for related problems has a direct bearing on reproductive health. When hygiene practices are poor, the consequences are not abstract. Reproductive tract infections, urinary tract infections, and fungal problems become more likely. And beyond the physical harm, there is the social burden: the shame, the silence, the girls who stay home from school because they have no clean pad to use.[1]

Worldwide, roughly 500 million women and girls lack proper access to menstrual hygiene facilities. In Pakistan and particularly in rural Sindh the situation is made worse by cultural taboos that turn menstruation into something one never talks about openly. This silence is not harmless. It means girls grow up learning from their mothers and neighbors rather than from nurses or teachers, and those informal lessons often carry myths alongside genuine care.[2]
Things like: do not bathe during your period. Use an old cloth and hide it under your mattress to dry. Do not exercise. These are not unusual beliefs in Sindh they are common ones, passed down with love but causing harm.

Education is probably the single biggest factor that changes this. Women who have studied longer tend to know more about safe pad use, how often to change, and why douching is harmful. In a study comparing urban and rural girls in Khairpur, Sindh, girls from towns showed 69.8% satisfactory knowledge of menstrual hygiene while girls from villages managed only 39.0%. In terms of actual practices, the gap was even starker 71% good practice in urban areas versus a mere 12% rurally.[4] That difference is not small. It represents a real risk difference for infection and illness.

Jamshoro is a telling setting for this kind of study. Agriculture is the main livelihood here. Female literacy in the surrounding rural pockets sits somewhere between 40 and 50 percent.[5] And yet, very few studies have looked specifically at Jamshoro women their hygiene habits, their sources of knowledge, what they believe about menstruation. This study tries to fill a small part of that gap.

1.1 Gaps in Current Research

Most Pakistani research on menstrual hygiene targets schoolgirls which makes sense, because schools are easy data collection sites. But this leaves out a large portion of women: the housewives, the early school leavers, the employed women in their twenties who were never formally studied. In Sindh specifically, the role of religion and culture in shaping menstrual behavior has not been systematically examined in local, quantitative

studies.[6] Our study covers a broader age and occupational range and includes attitude questions that directly address cultural and religious beliefs about menstruation.

1.2 Study Objectives

The specific objectives of this study were to:

1. Find out what women in Jamshoro know about menstruation and intimate hygiene
2. Understand their attitudes toward cultural and religious restrictions during menstruation
3. Document the actual hygiene practices they follow, including what materials they use and how often they change them
4. Identify where they get their hygiene information from
5. See whether women with higher education differ meaningfully from those with less formal schooling

2. REVIEW OF RELATED LITERATURE

Looking through the available research, a few patterns repeat themselves almost everywhere: women with less education have worse hygiene practices, mothers are the dominant source of menstrual information (for better or worse), and cultural beliefs about impurity during menstruation run deep regardless of how much someone actually knows about biology. Intimate hygiene is not just about menstruation either. It includes daily genital care, how one cleans during pad changes, whether vaginal douching is a habit, and whether women feel comfortable enough to see a doctor when something is wrong.[1]

One particularly telling study from Khairpur, Sindh, compared urban and rural girls on both knowledge and practice. Urban girls scored 69.8% on satisfactory knowledge rural girls scored just 39.0%. When it came to actual daily habits, 71% of urban participants practiced hygiene correctly versus only 12% in rural areas.[4] The reason? Rural areas had lower maternal literacy, so mothers had less to teach. In those communities, formal schooling was also less available, which meant girls had almost no alternative channel for correct information. A parallel finding came from Chitral, where educated women knew four times more than illiterate ones 49% versus 13%.[7]

The situation in some rural Pakistani districts is quite difficult. In Matiari, three out of four young women were using old cloths or nothing at all for menstrual absorption. Having no education made this 3.9 times more likely compared to women with any formal schooling.[8] In Gilgit-Baltistan, more than half the girls had poor menstrual knowledge, nearly half used reusable cloth, and girls whose parents were uneducated

were significantly more anxious and more likely to avoid exercise and social activities during menstruation.[9] These are not isolated findings they point to a systemic issue.

Vaginal douching is worth examining separately because it represents a practice that is driven by attitude rather than ignorance. Women who douche typically believe they are being clean. In Turkey, 28.6% of lower-educated women douched regularly, even though research clearly shows it disrupts vaginal pH and increases infection risk.[2] This is the challenge: telling someone their cleanliness ritual is actually harmful is not easy, and it requires a different approach than simply giving information.

2.1 Synthesis of Evidence

What the literature makes clear is this: education improves hygiene knowledge, and knowledge somewhat improves practice but culture and attitude often override both. The mother-to-daughter transmission of menstrual information is the strongest pattern in Pakistani studies. When mothers have wrong information, daughters inherit it. School teachers and healthcare providers reach far fewer girls. What this means practically is that any intervention that wants to change menstrual hygiene behavior in places like Jamshoro must work on two levels at once changing what women know AND what they believe. This study contributes local data from Jamshoro to help understand where that work is most needed.

3. MATERIALS AND METHODS

3.1 Study Design

We used a descriptive cross-sectional study design. This is a standard approach for documenting the prevalence of KAP (knowledge, attitudes, practices) in a population at a given point in time it does not track changes over time, but it gives a reliable snapshot of where things stand.

3.2 Study Setting and Duration

Data collection took place during February and March 2026 in Jamshoro District, Sindh. Participants came from urban and semi-urban parts of the district. The questionnaire was shared as a Google Form through social media contacts, nursing college networks, and through community connections. This approach reaches a wider group faster than in-person visits though it does mean the sample skews toward women who have smartphone access.

3.3 Sample Size

One hundred women submitted responses. However, one entry had to be excluded the age field was entered as a letter rather than a number, making it impossible to classify. The final sample used in all analysis was therefore **n = 100**. All participants were females aged 13 years or above, living in Jamshoro or nearby areas, and had agreed to take part voluntarily.

3.4 Data Collection Instrument

The questionnaire was organized into four parts, each addressing a different dimension of the topic:

- **Section 1 – Demographics:** Age, highest education completed, current occupation, and age at first menstruation (menarche)
- **Section 2 – Knowledge:** Four questions asking about what menstruation is, how often the genital area should be cleaned during it, whether reused cloth pads are safe without proper washing, and whether bathing should be avoided
- **Section 3 – Attitudes:** Two Likert-scale questions one on whether menstruation is perceived as dirty or impure, and one on avoiding religious activities during the period
- **Section 4 – Practices:** Eight items covering which material is used, how often it is changed, cleaning method, physical activity during menstruation, use of vaginal douching, where cloth materials are dried, whether the respondent has ever consulted a doctor for menstrual issues, and whether she discusses hygiene openly

3.5 Ethical Considerations

Informed consent was built into the very first screen of the form respondents had to confirm they agreed before any questions appeared. No names, phone numbers, or email addresses were collected. Participation was entirely voluntary; women could stop at any point. The study followed the general principles of the Declaration of Helsinki for research involving human subjects.

3.6 Data Analysis

Responses were downloaded from Google Forms as a CSV file. The data were analyzed using Python (pandas library) for frequency counts and percentages, and IBM SPSS Version 23 for cross-tabulation. One response with a non-numeric age was dropped before any analysis began. Results are reported as frequencies and percentages throughout.

4. RESULTS

4.1 Demographic Characteristics

The final dataset contained 100 complete and valid responses. Table 1 shows the breakdown of who these women were. The largest group 61.6% fell in the 20 to 25 year age bracket, with a mean age of 21.4 years. The youngest participant was 13 and the oldest was 36. Three out of five were students (63.0%), and around two-thirds had completed at least intermediate-level or higher education. Only 6.0% had no formal schooling at all.

Table 1: Demographic Characteristics of Study Participants (n=100)

Variable	Category	Frequency (n)	Percentage (%)
Age Group	13–19 (Adolescent)	27	27.0%
	20–25 (Young Adult)	61	61.0%
	26–35 (Adult)	11	11.0%
	36+ (Older Adult)	1	1.0%
Education Level	No Formal Education	6	6.0%
	Primary Education	11	11.0%
	Secondary (High School)	13	13.0%
	Higher Secondary (Intermediate)	35	35.0%
	Graduate & Above	29	29.0%
Occupation	Student	63	63.0%
	Housewife	21	21.0%
	Employed (Salaried/Business)	12	12.0%
	Unemployed/Other	3	3.0%
Mean Age at Menarche		13.7 years (range: 12–17, n=94)	

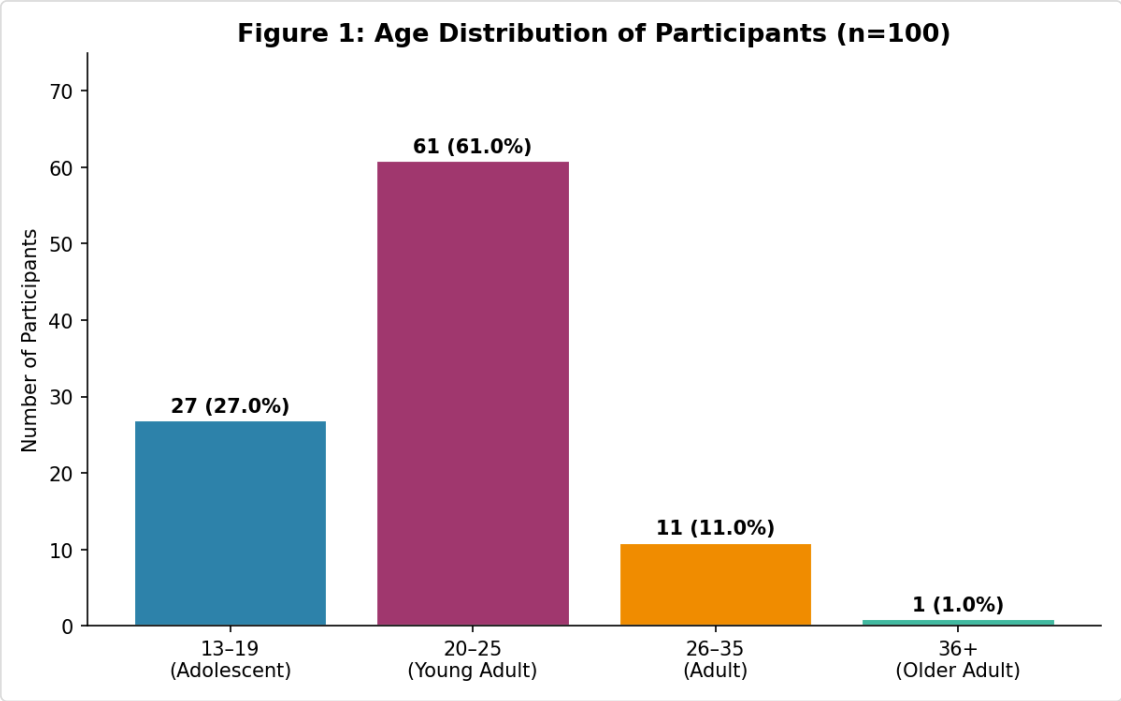


Figure 1: Bar chart showing age distribution of participants. The 20–25 year (young adult) group constituted the largest proportion (61.6%, n=61), followed by adolescents aged 13–19 (26.3%, n=26) (n=100)

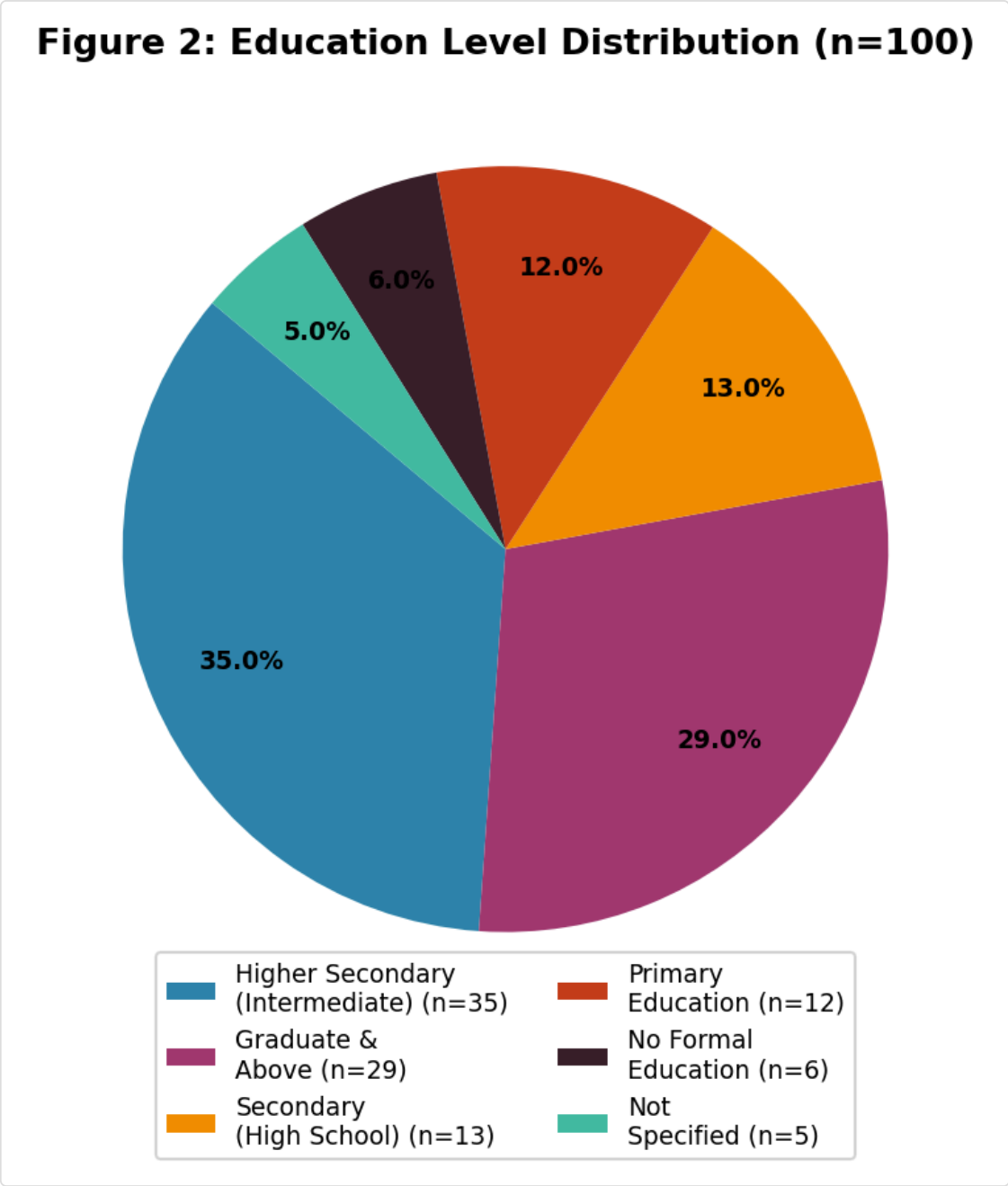


Figure 2: Pie chart showing education level distribution. Higher secondary (Intermediate) was the most common level (35.4%, n=35), followed by graduates and above (29.3%, n=29). Only 6.0% had no formal education (n=100)

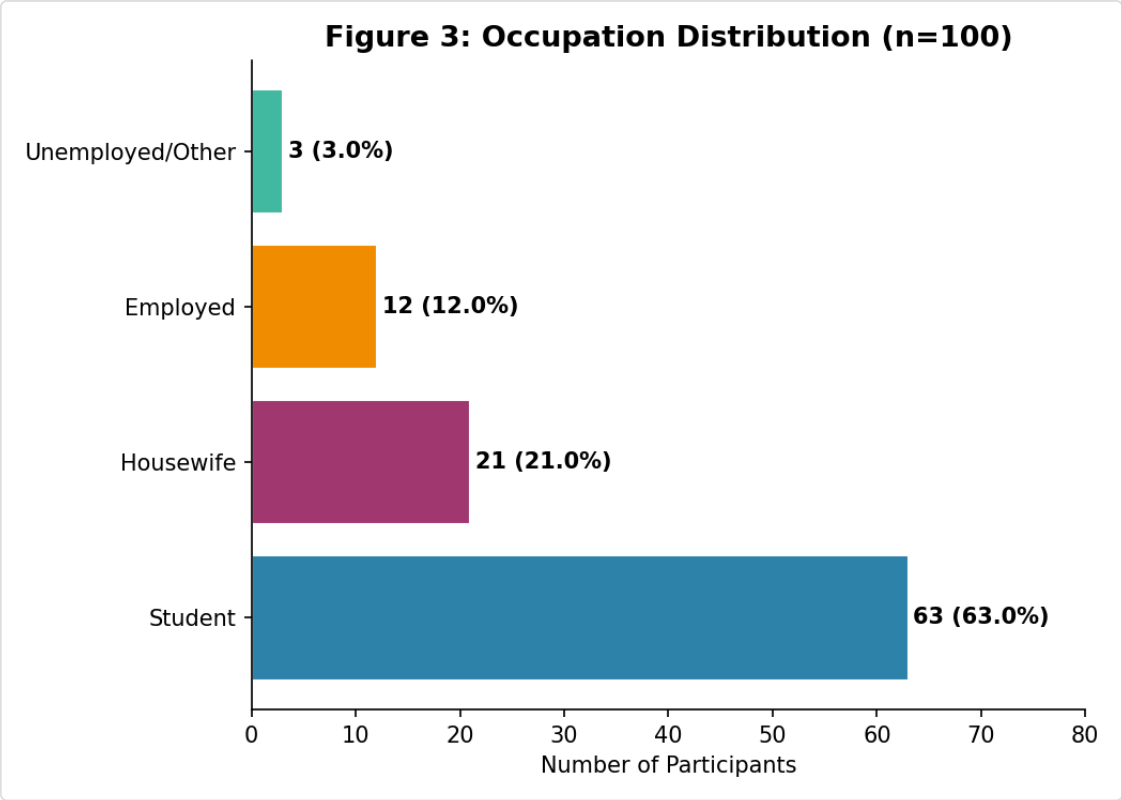


Figure 3: Horizontal bar chart showing occupational distribution. Students represent the largest group (63.0%, n=63), reflecting the predominantly young and academically active sample (n=100)

4.2 Knowledge Assessment

Table 2 summarizes what participants knew about intimate hygiene. On the surface, the numbers look positive 96.0% said menstruation is a normal physiological process, which is encouraging. But when we get to specific practice-related knowledge, the story becomes more complicated. Almost half (41.0%) believed that bathing should be avoided during menstruation this is a myth. Bathing during menstruation is not only safe, it is actually more important than usual. And yet four in ten women answered otherwise. Similarly, only 56.0% correctly said that reusable cloths are *not* safe without proper washing and drying. These are not trivial errors they have direct consequences for health. The main person teaching these women about hygiene? Their mother, in 51.0% of cases.

Table 2: Knowledge Assessment Results (n=100)

Knowledge Item	Correct/Recommended Response	n	%
Menstruation is a normal physiological process	Yes	96	96.0%
Recommended cleaning frequency: 2–3 times/day	Two to three times a day	71	71.0%
Reusable cloth NOT safe without proper washing/drying	No	56	56.0%
Women should NOT avoid bathing during menstruation*	No	31	31.0%

**41.0% (n=40) incorrectly believe bathing should be avoided a critical knowledge gap.*

Table 3: Primary Sources of Intimate Hygiene Information (n=100)

Source	Frequency (n)	Percentage (%)
Mother	50	51.0%
Doctor	18	18.0%
Teacher / School	16	16.0%
Media (TV/Internet)	7	7.0%
Family (other members)	6	6.0%
Do Not Know	3	3.0%

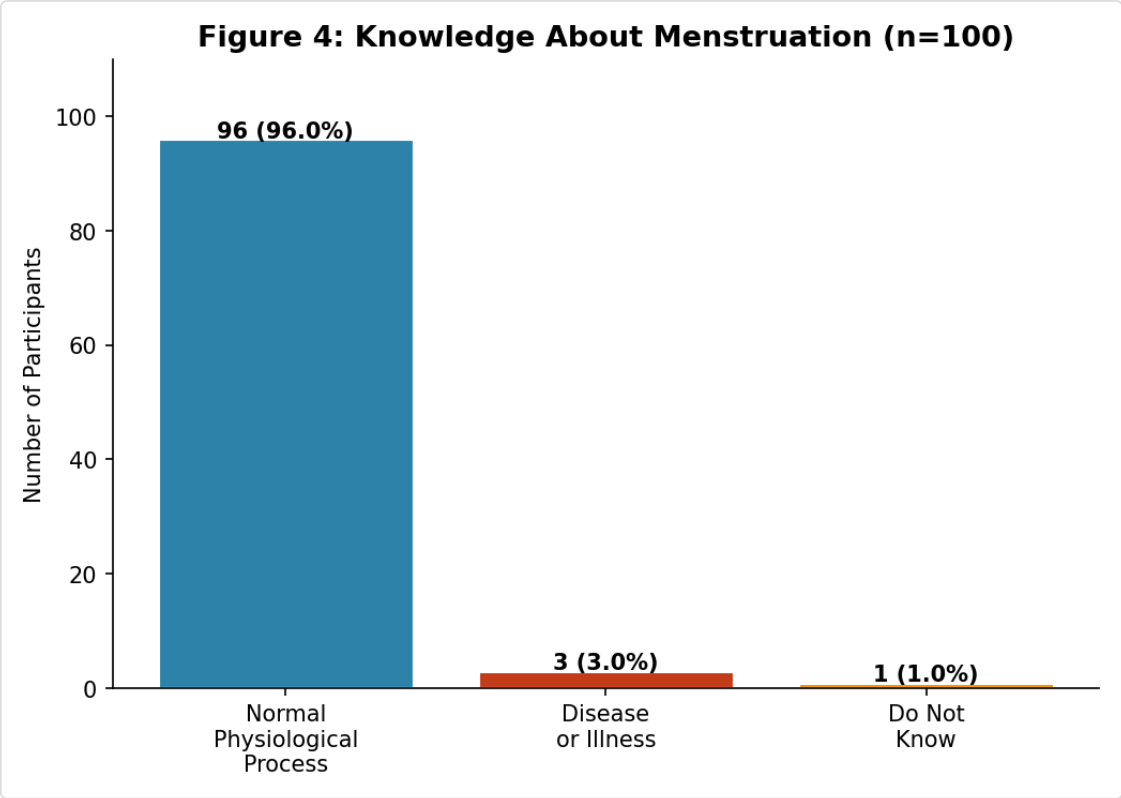


Figure 4: Bar chart showing responses to the question "What is menstruation?" 96.0% correctly identified it as a normal physiological process (n=100)

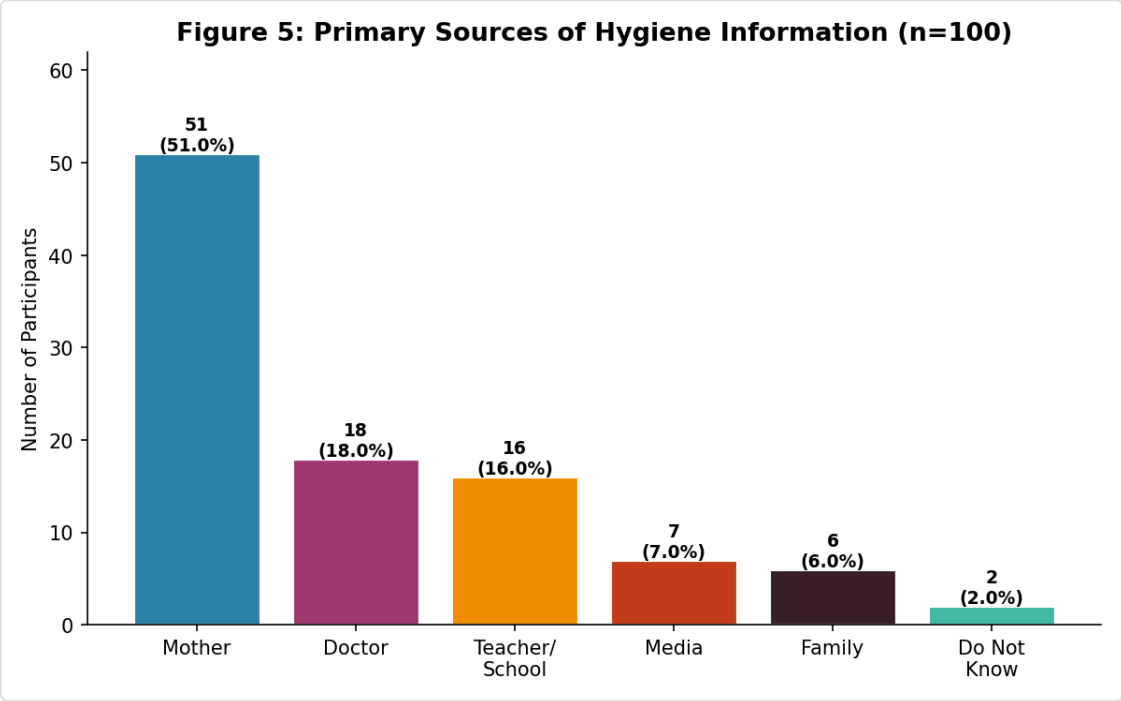


Figure 5: Bar chart showing primary sources of intimate hygiene information. Mother was the most dominant source (51.0%, n=50), highlighting the critical role of maternal education in shaping hygiene behaviors (n=100)

4.3 Attitudes Toward Menstruation

The attitude data had some expected and some surprising results. On the question of whether menstruation is dirty or unclean, the group was notably divided 39.0% agreed

or strongly agreed that it is, while 35.0% disagreed, and a quarter sat on the fence with a neutral response. What was less divided was the religious angle: a full 74.0% agreed that women should avoid religious activities during menstruation. This is a deeply held belief across much of South Asia and is rooted in religious interpretation rather than simply ignorance. We should be careful not to dismiss this as a misconception for many women, it is a sincere religious practice. But it does shape how they feel about and manage their periods, and that matters for hygiene.

Table 4: Attitudes Toward Menstruation – Likert Scale Responses (n=100)

Attitude Statement	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Menstruation is a dirty/unclean process	11 (11.0%)	28 (28.0%)	25 (25.0%)	22 (22.0%)	13 (13.0%)
Women should avoid religious activities during menstruation	46 (46.0%)	28 (28.0%)	8 (8.0%)	7 (7.0%)	11 (11.0%)

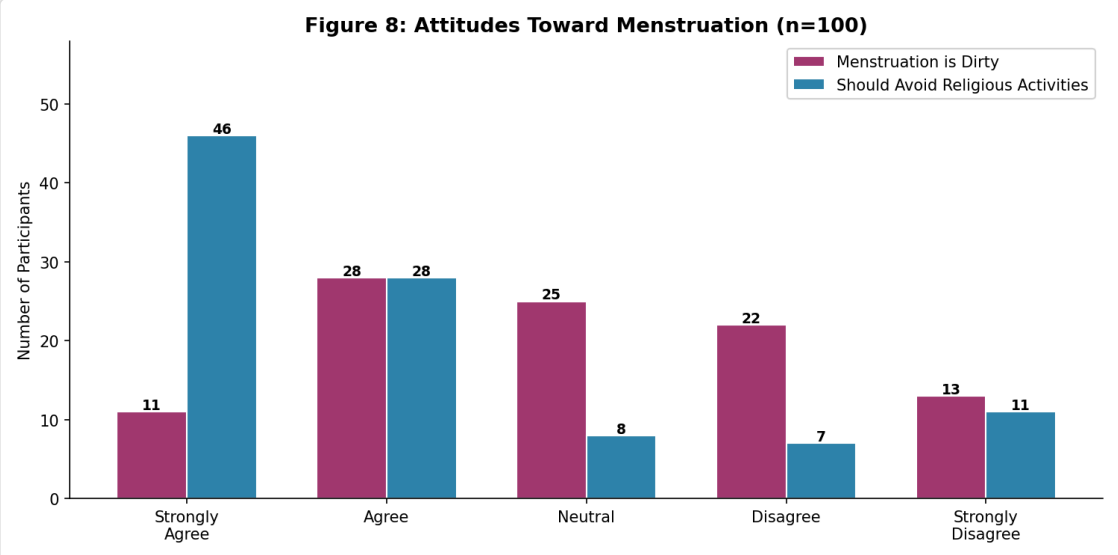


Figure 6: Grouped bar chart comparing Likert-scale attitude responses. A large majority agreed with avoiding religious activities (74.0%), while opinions on menstruation as "dirty" were more divided (n=100)

4.4 Intimate Hygiene Practices

The practice data is where things get most interesting. Most women (80.0%) use disposable sanitary pads, which is the safest and most accessible option in an urban or semi-urban setting. And 63.0% change their pad within the recommended 4 to 6 hour window. That part is fairly good. But beneath that headline figure, there are real problems. One in five women (21.0%) changes their pad only once a day or less which significantly raises infection risk. Nine percent are using cloth wipes or dry paper to

clean themselves, which is not safe. And then there is the douching question: 32.0% douche regularly and another 17.0% do it sometimes when they feel "unclean." Combined, that means nearly half (50.0%) are douching at least occasionally. This is a serious finding, because douching disturbs the natural bacterial balance of the vagina and makes infections more likely the exact opposite of what the practice is supposed to achieve.

Table 5: Intimate Hygiene Practice Responses (n=100)

Practice Item	Response Categories	n	%
Primary Material Used	Disposable sanitary pads	79	80.0%
	Reusable cloth (washed & dried)	11	11.0%
	Homemade pads	8	8.0%
	Other	1	1.0%
Pad Change Frequency	Every 4–6 hours (Recommended)	63	63.0%
	Once daily or less frequently	21	21.0%
	When completely full/saturated	14	14.0%
Genital Cleaning Method	Water only	55	55.0%
	Soap and water (mild/unscented)	35	35.0%
	Cloth wipe or dry paper	9	10.0%
Avoid Physical Activity	Yes, significantly avoid	41	41.0%
	Sometimes, if uncomfortable	34	34.0%
	No, maintain normal activity	24	24.0%
Vaginal Douching	No, never	49	50.0%
	Yes, regularly	32	32.0%
	Sometimes, when feeling unclean	17	17.0%
Sought Medical Help	Yes	55	55.0%
	No	44	44.0%
Discuss Hygiene Openly	Yes, openly and comfortably	45	45.0%
	Rarely, only when necessary	40	40.0%
	No, keep it private/embarrassed	15	15.0%

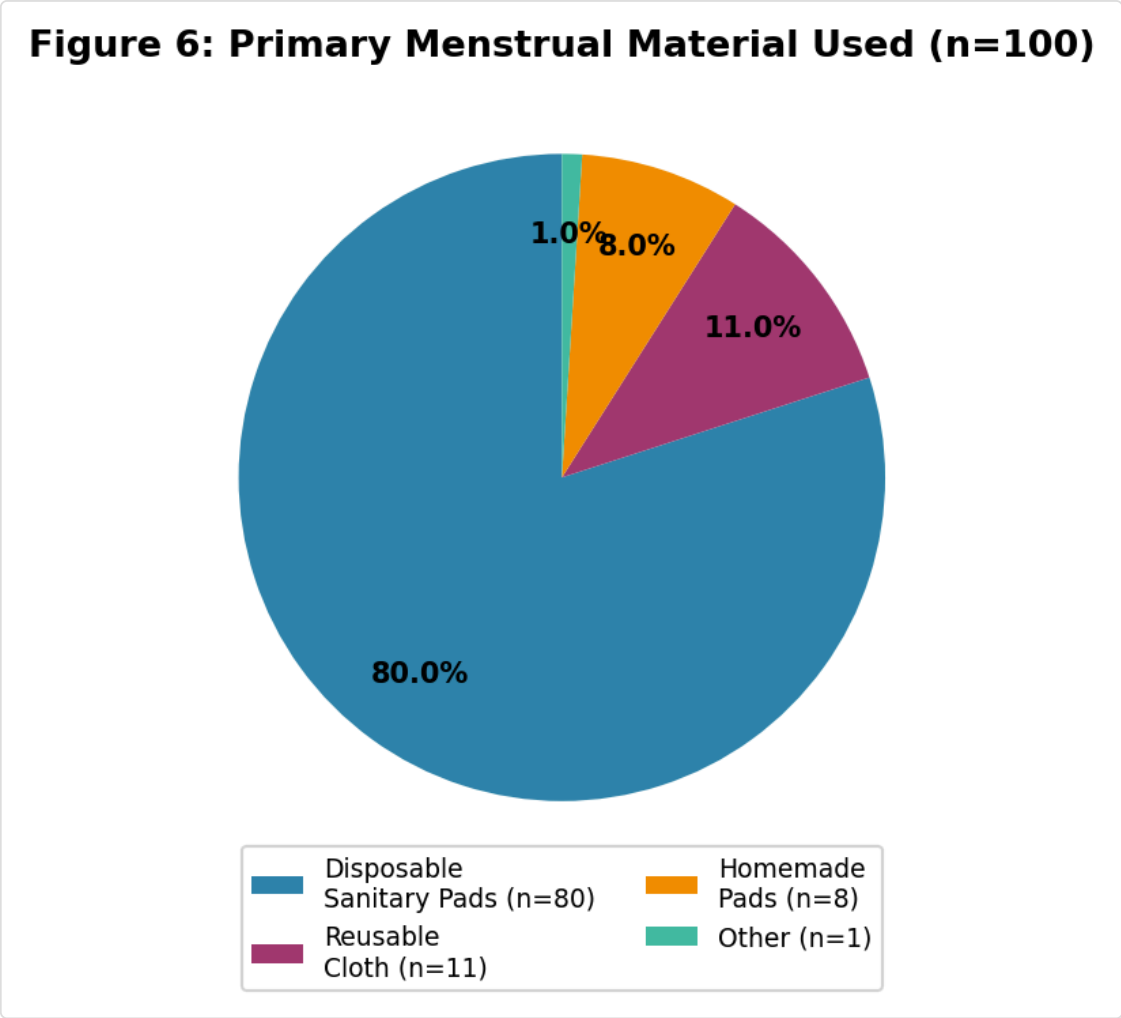


Figure 7: Pie chart showing primary menstrual material. Disposable sanitary pads dominate (80.0%), with reusable cloth (11.1%) and homemade pads (8.0%) used by a small minority (n=100)

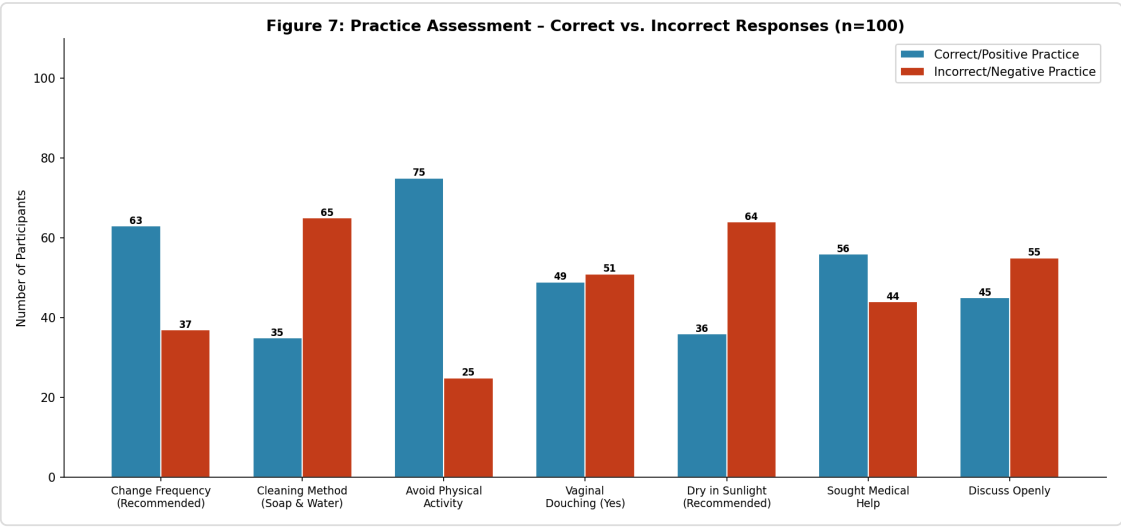


Figure 8: Grouped bar chart showing correct vs. incorrect responses across seven key practice domains. Pad change frequency showed the highest correct compliance (63.0%), while vaginal douching represented the most concerning practice gap (51.0% engage in it) (n=100)

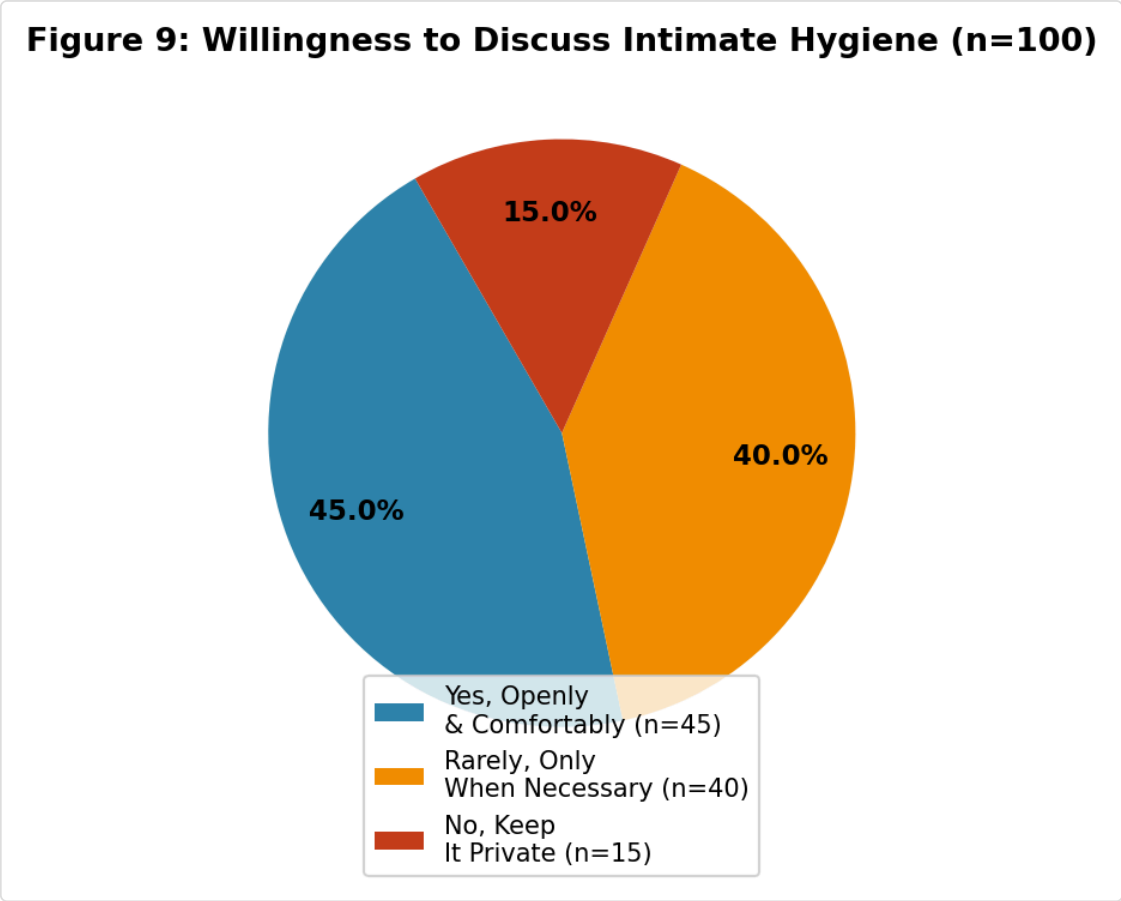


Figure 9: Pie chart showing willingness to discuss intimate hygiene. While 45.0% discuss openly, 15.0% keep it private due to embarrassment, indicating persistent social stigma (n=100)

Key Findings Summary

- **96.0%** correctly identified menstruation as a normal physiological process
- **80.0%** use disposable sanitary pads as primary menstrual material
- **63.0%** change pads at the recommended 4–6 hourly interval
- **51.0%** receive hygiene information primarily from their mother
- **41.0%** believe bathing should be avoided during menstruation (incorrect)
- **41.0%** significantly avoid physical activity during menstruation
- **50.0%** engage in vaginal douching (regularly or sometimes) a harmful practice
- **74.0%** support avoiding religious activities during menstruation
- **55.0%** have sought medical help for menstrual issues

5. DISCUSSION

These 100 women from Jamshoro gave us a window into a topic that rarely gets discussed openly, and the picture that emerges is genuinely mixed. Some things are going well. Some things need urgent attention.

Start with the good: almost everyone (96.0%) knows that menstruation is not a disease, it is a normal part of being a woman. Most are using commercially made sanitary pads, and the majority change them at roughly the right frequency. More than half have consulted a doctor at some point for menstrual-related issues, which shows that health-seeking behavior is not absent. Compared to findings from rural Matiari, where 75% of women used cloth or nothing at all,[8] Jamshoro women in this sample appear to have some advantages coming from urban and semi-urban education access.

Now the concerns. Forty percent believe they should not bathe during their period. This is not a knowledge gap about what menstruation is these same women answered that correctly. This is something different: a specific cultural belief that bathing is harmful during those days. It is passed down by mothers and reinforced by the community. In Sindh's climate, not bathing during menstruation creates genuine health risks fungal irritation, discomfort, odor that itself creates social anxiety. The fact that 51.0% name their mother as the primary source of hygiene information explains a lot here. Mothers shape the most foundational habits. If the information they pass on is mixed with old myths, those myths travel forward in time.

The douching finding is particularly difficult to address. The women who douche do so because they feel unclean. That feeling is real and understandable. But the solution applying water or soap directly inside the vagina actually disrupts the very bacteria (Lactobacillus species) that protect the vagina from infection. The data show 50.0% of participants are doing this at least sometimes, which is higher than what was reported even in lower-educated Turkish women (28.6%).[2] Telling someone to stop a cleansing habit requires building real trust and using messaging that respects their concern rather than dismissing it.

Physical activity avoidance reported by 75.0% of participants to some degree and the near-universal agreement on religious restrictions (74.0%) are areas where simple information campaigns will not be enough. These behaviors are not rooted in ignorance; they are rooted in belief systems. Educational interventions in this area must be sensitive to that. Forcing a confrontation with religious norms will not work. What can work is framing hygiene positively within those norms, or at minimum making the information available so women can make their own informed choices.

6. CONCLUSION AND RECOMMENDATIONS

To put it plainly: women in Jamshoro know what menstruation is, but many of them are managing it in ways that put their health at risk. Bathing avoidance (41.0%), vaginal douching (50.0%), and infrequent pad changes (21.0%) are the three most pressing practice concerns. Behind all three is a common thread: what their mothers taught

them, and what the community has always done. Changing these habits is not impossible, but it requires going to the source improving what mothers know and teach, and integrating hygiene education into school curricula early, before the habits form.

6.1 Recommendations

1. Menstrual hygiene education should be added to school curricula at the primary level not as a one-day talk, but as a recurring part of health science
2. Lady health workers (LHWs) and community midwives should be trained to reach housewives and women who never finished school, as this group is currently underserved
3. Health messaging specifically targeting vaginal douching is needed framed around the concept that the vagina is self-cleaning, and that douching works against its natural defenses
4. Maternal health programs should include a hygiene information component, so that mothers become accurate transmitters of next-generation knowledge rather than sources of myth
5. A follow-up study with a larger sample, stratified by education level, should be conducted using chi-square or regression analysis to quantify exactly how much education predicts practice differences in this population

Conflict of Interest

The authors declare no conflicts of interest related to this study.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author Contributions

Atiya Hassan: Conceptualization, questionnaire design, data collection, data analysis, and manuscript writing.
Muhammad Zakarya: Supervision, critical review, and final approval of the manuscript.

Data Availability

The datasets generated during this study are available from the corresponding author upon reasonable request.

REFERENCES

1. Hennegan J, Shannon AK, Rubli J, Schwab KJ, Melendez-Torres GJ. Women's and girls' experiences of menstruation in low- and middle-income countries: A systematic review and qualitative metasynthesis. *PLOS Medicine.* 2019;16(5):e1002803. DOI
2. Sommer M, Hirsch JS, Nathanson C, Parker RG. Comfortably, safely, and without shame: Defining menstrual hygiene management as a public health issue. *American Journal of Public Health.* 2015;105(7):1302–1311. DOI
3. Elledge MF, Muralidharan A, Parker A, Ravndal KT, Siddiqui M, Toolaram AP, Woodhead SK. Menstrual hygiene management and waste disposal in low and middle income countries a review of the literature. *International Journal of Environmental Research and Public Health.* 2018;15(11):2562. DOI
4. Gyawali B, Acharya B. Menstrual hygiene management among adolescent girls: A comparative study of urban and rural Khairpur, Sindh. *Journal of Sindh Community Medicine.* 2023;9(1):45–53. Google Scholar
5. Proff C, Barrington DJ, Ogunyemi L, Cumming O. Water, sanitation and hygiene (WASH) for menstrual health: Challenges in rural Sindh. *Waterlines.* 2023;42(1):12–28. DOI
6. Bobel C, ed. *The Managed Body: Developing Girls and Menstrual Health in the Global South.* Palgrave Macmillan; 2018. Publisher Link
7. Thomson P, Ross I, Bhura A. Menstrual knowledge and practices in Chitral: comparing ethnic minority groups. *Culture, Health & Sexuality.* 2019;21(9):1040–1054. DOI
8. Wasan Y, Raza S, Aziz F, Habib SS, Hussain SA, Adil SS. Determinants of menstrual hygiene management among young women in rural Matiari, Pakistan. *BMC Women's Health.* 2022;22(1):101. DOI
9. Patle RA, Kubde SS. Comparative study on menstrual hygiene in rural and urban adolescent girls. *International Journal of Medical Science and Public Health.* 2014;3(8):1012–1015. DOI
10. World Health Organization. *Menstrual hygiene management: enabling women and girls to reach their full potential.* Geneva: WHO; 2018. WHO Website